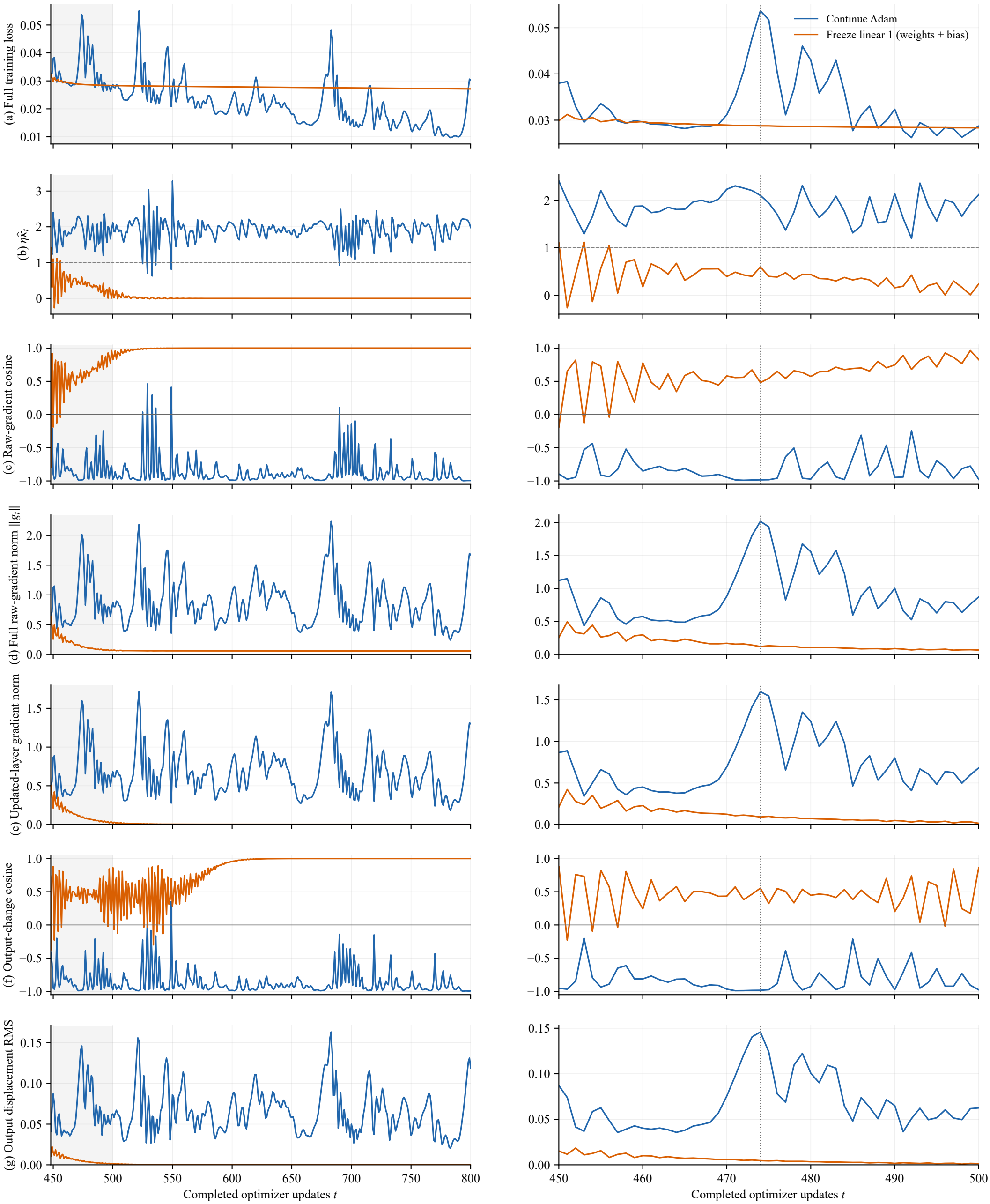
Freeze the first linear layer; keep Adam and the remaining learning rates unchanged

Fork at state 448; all 5,000 fixed training examples; LR = 0.001. No sample averaging or smoothing.

Full follow-up: 448-800

Event window: 450-500



$\hat{\eta\kappa}_t = 1 - \langle g_t, g_{t-1}\rangle / \|g_{t-1}\|^2$: projected-gradient diagnostic under Adam, not a curvature threshold.
Full gradients include all 656,810 parameters; row (e) uses only the 42,210 still updated. Output changes are outgoing.

Does the downstream network keep oscillating when the first representation is fixed?

Continue Adam
Freeze linear 1 (weights + bias)

Consecutive representation-change cosine
Displacement coordinate RMS
Cumulative singular energy

Linear 1 (Z_1)
Tanh 1 (H_1)
Linear 2 (Z_2)
Tanh 2 (H_2)
Linear 3 (F)

Frozen: zero motion
cosine undefined

Frozen: zero energy
normalized spectrum undefined

Outgoing update t

Number of fixed SVD modes

Each stream has its own uncentered SVD fitted on 450-500, retaining all sample-feature coordinates.
Concentrated singular energy remains after damping; it does not by itself diagnose oscillation or a stationary local axis.